LOGISTICS PRIMER

BITS GOA
CONSULTING CLUB

An overview of the global logistics ecosystem, covering key market dynamics, cost structures, value-chain players, and emerging trends shaping the industry.

This primer equips readers with concise insights on sector economics, growth drivers, and strategic challenges to help decode one of the world's most critical industries.

ACKNOWLEDGEMENT

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We extend our sincere gratitude to everyone who contributed to the creation of this Industry Primer. Your dedication, careful research, and thoughtful inputs have shaped this document into a resource that will support learners for many cohorts to come.

We thank the core contributors Aryan Gupta, Ayushmaan Singh, Chaitanya Mishra, and Samyak Patel, whose consistent effort, clear writing, and attention to detail strengthened the overall coherence and quality of this primer. Their work brought clarity and structure to each section and ensured that the content remained both accurate and accessible.

This edition has been developed entirely by BGCC and reflects the collective commitment of a team that worked with focus, discipline, and a shared sense of purpose. The contributions of each member added depth and helped create a well rounded and reliable reference. The collaborative spirit shown throughout the process has enriched the final outcome and allowed the primer to reflect diverse insights and perspectives.

We also express our deep appreciation to the leadership team Aashi Sengar, Harsh Vardhan, and Vishnuchittan Ramesh. Their guidance, coordination, and steady direction were central to the successful completion of this project and to maintaining the overall vision of the primer. Their efforts ensured that every stage progressed smoothly and that the document remained aligned with its intended purpose.

ABOUT BGCC

"Strategise. Strive. Soar."

BITS Goa Consulting Club (BGCC) is a student-led consulting group at BITS Pilani, Goa Campus. Since its establishment in 2020, the club has grown to 56 members who collaborate on research, analysis, and problem-solving for real businesses.

BGCC works closely with corporates, startups, non-profits, and public institutions, taking on live consulting projects across sectors. These engagements involve market research, growth strategy, financial and operational analysis, and other structured consulting work.

Over the years, the club has completed 75+ industry projects and partnered with 40+ clients, helping students gain practical exposure while supporting organizations with thoughtful, data-driven insights.

Thank you all for your contributions. This primer stands not only as a completed project but as a testament to what a motivated team can achieve together. We hope it continues to benefit future cohorts and inspires the same spirit of curiosity and collaboration with which it was created.

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INDUSTRY OVERVIEW

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TIMELINE

Before ~1800 CE: Ancient road and river networks evolved, by the medieval period, into efficient and sophisticated trading routes on both land and sea. Cargo would be loaded and unloaded from docked ships piece-by-piece manually.

1800 -1900 CE: The industrial revolution resulted in the creation of steam powered ships, followed by railways, which eased transportation. The Suez Canal and Trans-Siberian Railway were major milestones achieved for logistics during this period.

1900-1950 CE: The proliferation of modern road infrastructure and the birth of aviation contributed to making logistics faster. However, the problem of inefficient handling of cargo persisted. The Panama Canal was completed in this period.

1950-1980s CE: Aviation matured, and the creation of the standard cargo container in 1956 eliminated major inefficiencies in transporting cargo. European and Asian adoption of containerisation came a bit later, which made their ports more efficient.

1980s CE-Present: Technology has evolved, becoming a major part of logistics, going from spreadsheets towards AI. Roads, railways, waterways and air transport have matured into making the logistics industry indispensable in the modern world.

PESTEL ANALYSIS

POLITICAL: Shifting trade policies and FTAs impact global logistics through tariffs, regulations, and government spending shifts.

ECONOMIC: Freight demand closely tracks economic growth; logistics firms face risks from currency swings and fuel price volatility.

SOCIAL: Rapid e-commerce growth and instant delivery trends are redefining logistics with faster, consumer-driven models.

TECHNOLO-GICAL: Automation, AI, and predictive data analytics are continuously revolutionizing logistics efficiency and route optimization.

ENVIRONM-ENTAL: As a major global emitter, the industry faces rising pressure to cut carbon and adopt sustainable green practices.

LEGAL: Stricter customs, data privacy, and complex cargo compliance laws demand stronger corporate regulatory adherence.

KEY GROWTH DRIVERS

Technological Advancements: Ensuring competitiveness through a technological edge is essential for all global logistics players due to the heavy application of technology in this industry. Automotive processes for warehouse management are being widely adopted, ensuring that margins remain stable in unfavourable environments.

Increasing Consolidation: The industry is witnessing an increasing level of M&A activity due to the industry having immense economies of scale. M&A helps players merge their networks, making maritime shipping and warehousing more efficient. Heavy investments into technology are made easier with consolidation. Such factors help in the creation of synergies.

E-commerce and Globalisation: End market demand is fueled mostly by e-commerce businesses' distribution needs, which are fulfilled by logistics companies. Logistics, being a crucial component of the global e-commerce supply chain, is positioned well to benefit from the growth of e-commerce, which has a staggering market size CAGR of 18.9%.

MARKET SIZE

	AMERICAS	EUROPE	ASIA
Share of global market	33%	20%	35%
Spending as % of GDP	9%	8.5%	12%
	$1.30T	$780B	$1.38T
	CAGR: 2%	7.3%	7.6%

INDUSTRY OVERVIEW

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INDUSTRY DYNAMICS & SYSTEM FRICTIONS

Value Creation Through Movement : Creates economic value by converting distance & time into service - paid through freight rates, storage fees, and handling charges.

Multimodal Revenue Streams: Money is earned across various modes via transport charges, container fees, terminal handling, warehousing and last-mile delivery.

High-Volume, Low-Margin Economics: Relies on high throughput and strong asset utilization; margins improve only when vehicles run fuller and turn faster.

Cost Sensitivity & Turnaround Time: Fuel, labor, equipment, dwell time, and congestion directly influence cost per shipment; reducing turnaround time lifts yield per asset.

Efficiency Through Digital Visibility: Digital systems like ULIP, LDB, reduce documentation delays, cut empty runs, and smooth cash cycles - raising overall operating margins.

VALUE LEAKAGES

- Fragmented Planning
- Port & Corridor Bottlenecks
- Rail Capacity Constraints
- Handling & Storage Gaps
- Visibility & Compliance Delays

KEY TERMS

Cross-Docking: Transferring goods directly from one vehicle to another with minimal to no storage in between.

Common Carrier: A transport company that offers services of moving goods for the public at published rates.

FOB (Free On Board): Seller delivers goods on board the ship, following which the buyer takes the risk and cost.

CIF (Cost, Insurance & Freight): The seller pays cost, insurance, and freight up to the destination port.

DDP (Delivered Duty Paid): Seller handles all costs including customs and transport, up to buyer's doorstep.

TEU/FEU: Mean twenty-foot equivalent unit and forty-foot equivalent unit, standard units of measuring containers.

WMS (Warehouse Management System): It is a software that controls warehouse processes like tracking.

Green Corridors: Routes optimized for low carbon emissions via alternative fuels or efficiency measures.

KEY GOVERNMENT INITIATIVES

National Logistics Policy

A national framework to reduce logistics cost, streamline processes, and build a unified, future-ready supply-chain ecosystem across India.
Strengthens digital governance through ULIP and the Logistics Data Bank, enabling real-time visibility, transparency, and smarter data-driven planning.

PM GatiShakti Master Plan

A GIS enabled platform integrating 57 ministries, 36 states and 1700+ data layers to accelerate coordinated multimodal infrastructure development.
Improves corridor alignment, avoids duplication, reduces execution delays, and ensures seamless long-term connectivity across different transport modes.

Dedicated Freight Corridors

High-capacity, freight-exclusive railway corridors designed to shift heavy cargo from road to rail for faster, cleaner, lower-cost movement.
Reduce congestion on passenger routes, improve energy efficiency, offer predictable transit schedules, and strengthen India's industrial logistics backbone.

Multi-Modal Logistics Parks

A network of 35 large-scale logistics hubs integrating warehousing, cold chain, transport and container handling in a unified modern ecosystem.
Enable smoother multimodal shifts, reduce handling time, support freight consolidation, and significantly lower overall logistics cost for industries.

Maritime Vision 2047, GST

Long-term roadmap to modernize ports, expand coastal shipping, digitize operations and build greener, more resilient maritime infrastructure.
GST and e-Way Bills simplify interstate movement, improve compliance, reduce checkpoints, and deliver faster, more transparent nationwide goods flow.

INDUSTRY OVERVIEW

SEGMENT COMPARSION MATRIX

	End User	Capital Intensity	Labour	Ticket Size	Tech Adoption	Revenue Model	Barriers to Entry
3PL	B2B	Medium-high	Medium	High	High	Contractual	Medium
Freight Forwarding	B2B	Low-medium	Medium-high	High	Medium	Contractual, Booking margins	Medium
Last-Mile Delivery	B2C	High	Very high	Low	High	Delivery Commissions	Medium
Q-Commerce	D2C	High	Very high	Very low	Very high	Delivery Commissions	Low
Warehousing	B2B/B2C	Very high	High	Medium	High	Rent, lease contracts	Medium
Cold Chain	B2B	Very high	Medium	Medium	Medium	Temperature and Time based	High
Road Freight	B2B/B2C	Low-medium	Low	Medium	Low	Quantity and distance based	Low
Rail Freight	B2B/B2C	Low-medium	Medium	High	Low	Quantity and distance based	High
Air cargo	B2B/B2C	High	Low	High	Medium	Weight and distance based	Medium
Waterways	B2B	High	Low	High	Low	Weight and Distance based	High
Reverse Logistics	B2B/B2C	Low-medium	Medium	Very low	High	Resale, handling fee	High

KEY RISKS

Disruption risks: Natural disasters, logistics malfunctions, strikes, and political upheaval can disrupt supply chains. A study revealed extreme weather delayed shipments by an average of two days and led to cancellations for up to 75% of scheduled shipments.

Financial risks: These primarily arise from volatile market conditions. Fluctuations in fuel prices, liquidity management, treasury risks, credit management and contract audits can significantly squeeze margins and affect profitability across supply networks.

Compliance risks: It can be difficult to adapt to customs regulations of different countries. Delays in customs procedures or lack of documents may prevent the products from being delivered on time which may lead to fines, operational disruptions or potential loss of licenses.

Operational risks: Equipment failure, including ageing infrastructure, insufficient maintenance, or unforeseen malfunctions, can cost companies up to a 54% decline in sales and a 46% decline in market share. Inefficient route planning can lead to increased fuel usage and wasted time.

FUTURE OUTLOOKS

Automation

Digital platforms, AI, and robotics streamline operations and improve efficiency. Autonomous vehicles and drones can revolutionize last-mile delivery last-mile delivery in urban and remote areas

Big Data & IoT

Businesses can gain valuable insights into consumer behavior, demand patterns, and supply chain performance. IoT-enabled logistics can identify inefficiencies and optimize inventory management.

Sustainability

Some key trends are adoption of alternative fuels and electric vehicles, use of renewable energy sources, such as solar and wind power, biodegradable packaging to reduce waste.

Policies

National Logistics Policy(NLP) focuses on cost reduction, efficiency, digitalization, and sustainability. Moreover, International policies emphasize harmonization and resilience.

SEGMENT-WISE ANALYSIS

With the overall logistics landscape established, the next section dives into each segment individually, unpacking market structure, key players, cost drivers, and emerging trends.

RAIL FREIGHT

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OVERVIEW

"India's logistics backbone is moving on rails — faster and greener.."

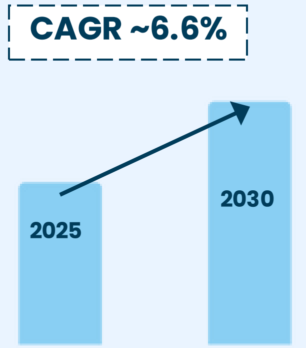
Definition & Scope : Rail freight transport involves the use of railways to move cargo, including bulk commodities (e.g., coal, iron ore), intermodal containers, general freight, and specialized goods, as opposed to human passengers. It is a vital component of the logistics chain, enabling long-distance, high-volume transport with significant environmental and economic benefits.

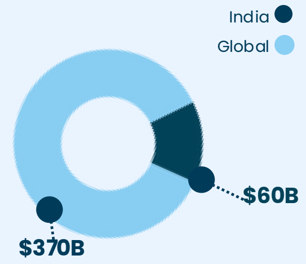
The Indian Scope: In India, rail freight accounts for approximately 30-36% of freight traffic (ton-km). Rail freight supports the India's ambition to become a US$26 trillion economy by reducing logistics costs, enhancing supply chain efficiency, and supporting industrial growth.

VALUE CHAIN

Origin & First-Mile →	Line-Haul Transport →	Terminal Operations →	Last-Mile Distribution →	Reverse Logistics
Cargo is collected from mines, factories, or ports and moved to the nearest rail terminal for loading.	Rail moves containerized cargo efficiently across main corridors and DFCs(Dedicated Freight Corridors)	At rail terminals, cargo is loaded/unloaded sorted, stored, and transferred between rail and road modes.	Goods are transported from destination terminals to distribution centers via road transport.	Empty wagons/containers are repositioned and return loads managed to reduce empty runs.

KEY NUMBERS





Global vs India

The Global Warehousing market expected to reach **$0.5T** by 2030

Indian Railways moves over **1.5 billion tonnes** of freight annually, making it one of the world's largest bulk transport systems.

COMPETITIVE LANDSCAPE

AMERICAS

BNSF RAILWAY · UNION PACIFIC · CSX · CN

EUROPE

DB Cargo · SNCF · PKPCARGO · ÖBB

ASIA

JR · KORAIL

MIDDLE EAST & AFRICA

SAR · ETIHAD RAIL

GROWTH DRIVERS

Infrastructure Expansion: Investments in Dedicated Freight Corridors (DFCs), terminals, and multimodal logistics parks are boosting capacity, speed, and network efficiency.

Cost & Modal Shift: Rail's lower long-haul costs and far lower emissions are pushing cargo movement away from road toward rail.

Intermodal Growth: Rising containerization and stronger port-rail-road links enable faster, more reliable movement of diverse cargo.

Industrial Demand: Steel, cement, coal, and manufacturing sectors drive high-volume, long-distance freight requirements suited to rail.

Digital & Private Push: Tracking systems, digital scheduling, and private terminals enhance transparency, efficiency, and service quality.

CHALLENGES

High Tariffs & Costs: Freight tariffs remain high compared to global benchmarks, affecting competitiveness against road transport.

Infrastructure Bottlenecks: Limited modern terminals, outdated handling equipment, and slow adoption of automation hinder efficiency.

Last-Mile Gaps: Insufficient terminal access and weak first/last-mile connectivity increase handling time and overall logistics cost.

Capacity & Congestion: Legacy corridors face heavy congestion, limiting speed, reliability, and available train paths for freight.

Service & Visibility Issues: Inconsistent tracking, delays, and limited visibility reduce service quality and customer confidence.

WATERWAYS

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"Waterways are powering India's low-cost, low-carbon logistics shift."

Definition & Scope : . Waterways logistics includes ocean shipping, coastal movement, and inland waterways transport that move bulk cargo, containers, liquids, and oversized goods through sea routes and navigable rivers. It is a cost-efficient and energy-efficient mode suited for high-volume, long-distance freight, reducing pressure on road and rail networks.

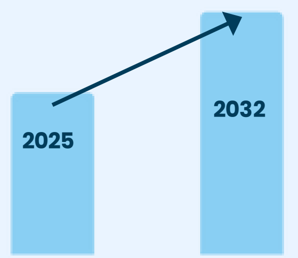
The Indian Scope: Ocean shipping handles 95% of India's trade by volume and 70% by value, while Inland Waterways Transport (IWT), despite its vast potential across 14,500 km of navigable waterways, accounts for only 0.5% of freight traffic.

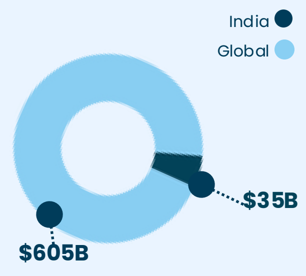
VALUE CHAIN

Cargo Aggregation	Port Handling	Waterway Transport	Inland Distribution	Reverse Logistics
Cargo is collected from factories, mines, or warehouses and moved to the nearest port or jetty for loading.	Cargo is received, stored, and loaded onto vessels or barges using cranes, conveyors, and handling equipment.	At rail terminals, cargo is loaded/unloaded sorted, stored, and transferred between rail and road modes.	Cargo is unloaded at ports or river terminals and delivered to warehouses or industries by road or rail.	Empty containers and barges are repositioned, and return loads managed to reduce empty trips.

KEY NUMBERS

CAGR ~6%





The Global Waterways Transport expected to reach **$960B** by 2032

Indian ports handle **700+ million tonnes** of cargo annually, reinforcing waterways as the backbone of India's EXIM trade.

COMPETITIVE LANDSCAPE

AMERICAS	EUROPE
ACBL, KIRBY, INGRAM, CMA CGM	RHENUS LOGISTICS, MSC, COSCO SHIPPING

ASIA	MIDDLE EAST & AFRICA
SCI, Trans Asia Shipping Lines, Inc., TCI Seaways	DP WORLD, Bahri, Bolloré Africa Logistics

GROWTH DRIVERS

Port & Waterway Development: Upgrades under Sagarmala, new river terminals, deeper drafts, and modern port infrastructure are boosting capacity and improving cargo flow.

Cost Advantage: Waterways offer the lowest cost per ton-km, making them ideal for bulk cargo and encouraging modal shift from road and rail.

Coastal & IWT Expansion: Growth in coastal shipping and inland waterways movement, supported by National Waterways and new barge routes, is increasing freight volumes.

Industrial Demand: Rising export-import flows and high demand from steel, cement, coal, and agriculture are increasing the need for water-based logistics.

Digital & PPP Push: Digital tracking, river information systems, and private participation in ports are improving efficiency and service reliability.

CHALLENGES

Draft & Depth Limitations: Shallow depths, siltation, and limited dredging restrict vessel size and reduce navigability on key waterways.

Port & Terminal Constraints: Limited berths, older equipment, and port congestion slow loading/unloading and lower efficiency.

Seasonal Navigability Issues: Fluctuating water levels cause seasonal closures, route limits, and inconsistent transit times.

Last-Mile & Hinterland Gaps: Weak port-to-hinterland links and poor multimodal integration increase handling time and costs.

Regulatory & Coordination Hurdles: Multiple authorities, slow clearances, and uneven state regulations create delays for operators.

AIR CARGO

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OVERVIEW

"Air cargo: speed and reach, worldwide."

Definition and Scope : Air Cargo refers to transporting goods via aircraft, offering the fastest way to move cargo across long distances. Businesses rely on air cargo for its speed, reliability, and global reach, making it essential for industries with time-sensitive, high-value, or perishable goods.

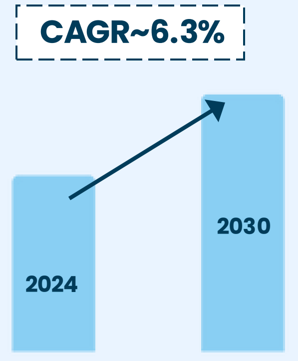
Sustainability test : The sustainability test for air cargo spans cost viability, efficiency, and adaptability. Despite unmatched speed, it faces pressures from fuel costs, regulations, and infrastructure limits. Investments in fleet upgrades, digital tracking, and cargo hubs are strengthening future readiness.

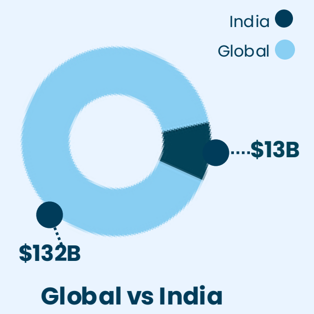
VALUE CHAIN

First-Mile & Booking → Origin Handling → Air Transit → Destination Handling → Final Delivery

First-Mile & Booking	Origin Handling	Air Transit	Destination Handling	Final Delivery
Cargo pickup and consolidation from shippers; capacity booking through digital freight forwarders and airline platforms.	Secure warehouse, customs export clearance, and cargo build-up into Unit Load Devices (ULDs) for flight readiness.	High-speed transit of goods using dedicated cargo freighters or passenger aircraft holds on scheduled routes.	ULD breakdown on arrival, import customs clearance, sorting of individual shipments at the destination airport warehouse.	Time-specific delivery from airport to consignee, with proof of delivery and cargo claims management.

KEY NUMBERS





Air cargo moves **less than 1%** of global trade **by weight**, yet carries over **35% by value**.

With **6–9% annual growth**, **India** is among the **fastest-expanding** air cargo markets globally.

MARKET SEGMENTATION

Based on **Service**

General
General cargo includes machinery, textiles, auto parts, and consumer goods moved on scheduled flights. It represents the majority of air freight, usually without strict delivery deadlines. It is slow but cost-effective.

Express
Express cargo covers urgent shipments such as e-commerce parcels, documents, and medical supplies. It is handled by integrators like DHL, FedEx, and Blue Dart with guaranteed speed and end-to-end tracking.

Based On **End-Use**

- **Pharmaceuticals & Healthcare**
- **Consumer Electronics**
- **Automotive**
- **Retail & E-Commerce**
- **Food & Beverages**

VALUE PROPOSITIONS

Reduced transit time: Air transport delivers faster, often within 24-48 hours. This can be crucial for time-sensitive items like highly perishable foods, certain pharmaceuticals, and high-value goods.

Global reach: Air freight can reach almost anywhere with its extensive route network, facilitating global trade.

Lower damage risks: Air transport comes with a reduced risk of theft and damage due to strict regulations and faster transit.

Improved inventory management: Faster deliveries ensure businesses maintain leaner inventory levels, supporting just-in-time inventories and reducing storage costs.

Humanitarian & emergency supplies – fast delivery of relief materials during crises.

CHALLENGES

High Costs driven by fuel prices, aircraft maintenance, airport fees, and the need for specialized labor makes it unviable for shipping low-value, heavy, or bulky goods.

Environmental Impact: Aviation has a large carbon footprint, emitting substantial greenhouse gases. This makes air cargo the least sustainable shipping option per ton.

Cargo Restrictions: Air freight is limited by size, weight, and hazardous cargo regulations. It is not suitable for transporting excessively large, bulky, or heavy shipments.

Operational Disruptions and Inflexibility: Air cargo is limited to an airport-to-airport network. It is also vulnerable to disruptions from adverse weather, causing significant delays.

Regulatory Hurdles: Stringent international and national regulations, including rigorous security screenings.

AIR CARGO

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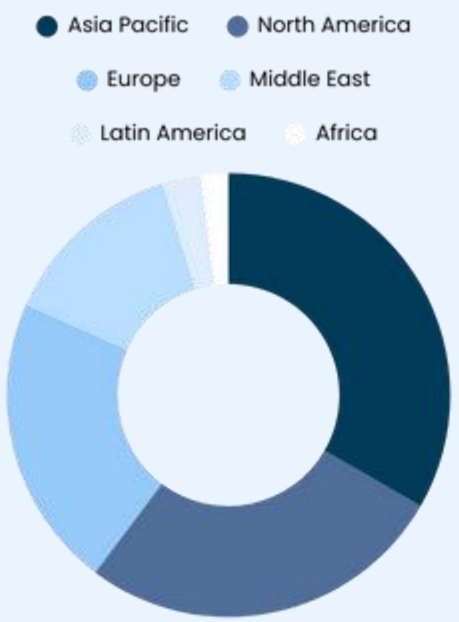
COMPETITIVE LANDSCAPE



Integrators

Ex: DHL, FedEx, UPS

- *Global express leaders manage ~40% of worldwide express cargo volumes.*
- *DHL Express revenue: €27B (2023); FedEx Express: $42B (2023).*
- *Strong in pharmaceuticals, e-commerce and time-critical shipments*

All-Cargo Freighter Operators

Ex: Atlas, Cargolux, Korean Air Cargo

- *Operate dedicated freighter fleets for oversized and bulk cargo.*
- *Cargolux revenue: $5.1B (2023); Atlas Air has ~100 freighters globally.*
- *Smaller market share than integrators, but essential in intercontinental lanes.*

Passenger-Carrier Belly Cargo

Ex: Emirates, Qatar, Cathay

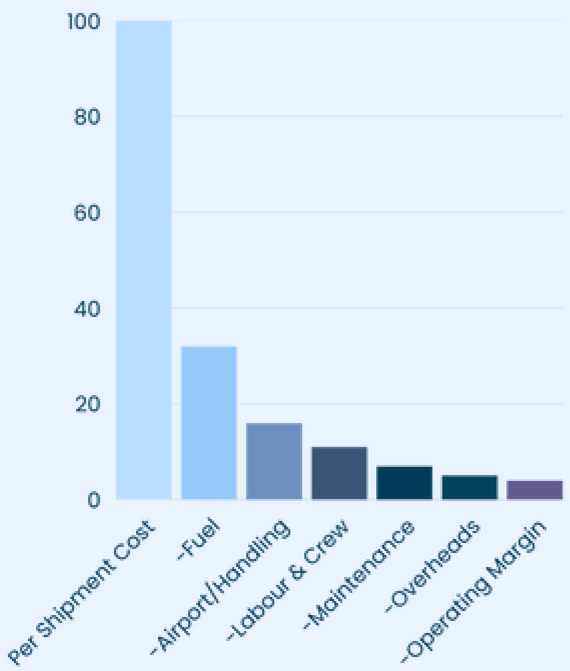
- *Belly space accounts for ~50% of global air cargo capacity.*
- *Emirates SkyCargo moved ~2.1M tonnes (2023); Qatar Airways Cargo ~2.0M tonnes.*
- *Performance linked to passenger traffic recovery and long-haul demand.*

India Specialists

Ex: Blue Dart, SpiceXpress, IndiGo Cargo

- *India managed 3.7 million tonnes in FY25, a 10% YoY increase.*
- *Blue Dart Express revenue: ₹5,450 Cr (2023).*
- *SpiceXpress demerger, valued at approx ₹2,500 Cr, while IndiGo is expanding its freighter fleet.*

PER-SHIPMENT ECONOMICS



Cost Pressures :
High unit costs, with fuel making ~30–35% of operating expenses and airport/handling charges adding another 15–20%. Yields are volatile, fluctuating with global demand and belly capacity availability.

Path to Breakeven:
Profit depends on high loads, route planning, and premium cargo (pharma, perishables, express). Margins are strongest on long-haul routes, while domestic runs thin with low volumes and higher costs.

ECONOMIC ENABLERS

Load Factor Efficiency: Higher payload utilization across freighters and passenger belly space reduces unit costs and improves profitability. Airlines target 75–85% load factors to keep operations viable.

Cargo Mix Value: Premium cargo (pharma, electronics) yields 20–30% higher revenues than general cargo, strengthening margins and reducing exposure to price swings.

Policy & Infrastructure Support: Schemes like Krishi UDAN 2.0 and airport cargo hubs cut operating charges and boost regional connectivity, supporting volume growth in India's market

CASE STUDY: BLUE DART

BLUE DART

"Blue Dart Express, backed by DHL, is India's top express air cargo carrier, supporting e-commerce and pharma supply chains while expanding its network with new aircraft, infrastructure, and digital solutions."

REVENUE TREND

- **1983 – Founding:** est. as a courier service; builds strong domestic express network.
- **2004 – DHL Alliance**: DHL invests €120M, alliance strengthens global connectivity.
- **2019 – Cargo Hub Expansion:** Opens major facility at Mumbai airport; fleet expanded and revenue crosses ₹3,100 Cr.
- **2021 – E-commerce Surge:** Revenues rise to ₹4,400 Cr, profits rebound
- **2023 – Fleet Growth:** Adds 2 Boeing 737, expands reach to 56,000+ locations, revenue at ₹5,172 Cr.

EXECUTION LEVERS

Integration with DHL
Access to DHL's global routes, tech, and infrastructure

Express Model
Time-definite cargo; growth in e-commerce, pharma, auto parts.

Tech & Ops
IoT-enabled cold chain, digital freight booking, and route optimization.

SIGNIFICANCE

- India's largest domestic air cargo.
- Backbone for e-commerce & pharma.
- Market leader in time-critical deliveries.

FINANCIAL HEALTH

- FY23 revenue: ₹5,172 Cr, PAT ₹366 Cr.
- Operating margin: ~9–10%.
- Daily throughput: 3.5M+ shipments

FUTURE OUTLOOK

- Expand freight and reach Tier-2/3 cities.
- Develop pharma cold chain and healthcare logistics.
- Scale EV fleet and green logistics hubs.

FREIGHT FORWARDING

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OVERVIEW

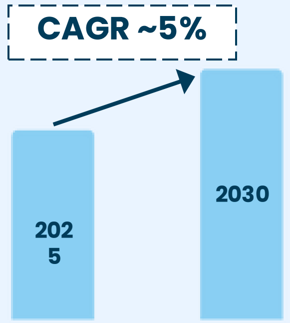
Freight forwarding: From a relay race in the dark to a glass pipeline

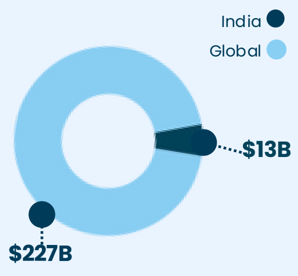
Freight Forwarders' Role as an Intermediary: Traditionally, the value that freight forwarder brought was that of an agent, arranging the movement of shipments in an opaque system. This would be done by organising the shipment's route through a network of ocean carriers, while handling all regulatory hurdles.
Visibility Increasing via Technology: Freight forwarders now, on top of their previously offered services provides intelligence that makes the complexity of shipping more manageable and transparent. Along with this, they also now offer data such as the most reliable trade lanes, carriers offering the best on-time performance and design routes that optimise for cost, speed and reliability.

VALUE CHAIN

Shipment Planning →	Export Handling →	International Transit →	Import and Customs →	Final Delivery
Shipments' routes are planned considering cargo, origin, destination, and time sensitivity	Cargo is booked with the selected carrier, while coordinating local pickup and warehousing.	The forwarder actively tracks the shipment, giving updates such as port delays or route changes.	Upon arrival, they coordinate with customs brokers to ensure compliance and release of goods.	The final leg of transport involves delivering goods from the port of entry to the client's recipient.

KEY NUMBERS





Global vs India

The Global Freight Forwarding market is expected to **1.3× in 5 years**

BUSINESS MODEL

Forwarders offer their expertise in customs regulations, documentation requirements and international trade laws, helping clients avoid costly delays and fines. By consolidating shipments from multiple clients, they get better freight rates from carriers. They handle cargo insurance, manage documenation and solve problems that arise during transit.

A primary source of profit for freight forwarders is a freight margin, wherein a rate is negotiated with a carrier and charge their clients a higher rate. This spread is their profit margin. Service fees are billed for specific activities like documentation, customs brokerage and shipment handling fees. Ancillary services such as cargo insurance and warehousing is also a revenue source..

KEY PLAYERS

Kuehne+Nagel: A global logistics company that provides sea freight and airfreight forwarding.

DHL Global Forwarding and Freight: It carries goods via road, rail, air or sea globally, under the DHL brand.

DSV: Global player offering freight forwarding across various transportation modes.

Nippon Express: It is a global logistics company, offering comprehensive freight forwarding services.

Sinotrans: It has a presnece in 32 countries and offers international freight forwarding.

MARKET DYNAMICS

The threat of new entrants into the industry is moderate as a major requirement is having a large network of carriers.

This notion is being changed by asset-light freight forwarders, using digital forwarding owing to greater technology use.

The competitive nature of the market is due to the large number of players, all offering very standardised services.

Competition has been intensifying among the top companies as consolidation to diversify services offered continues.

Example: CMA CGM, a major shipping company, acquiring Bollorè Logistics, a global freight forwarding company.

3PL

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OVERVIEW

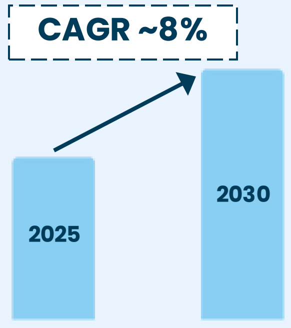
"The autonomous supply chain is rewriting the global 3PL playbook."

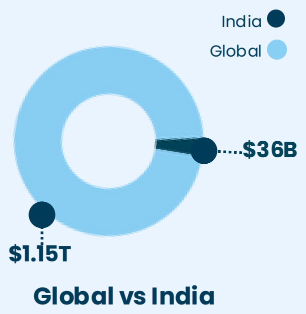
3PL (Third-party logistics) as an Executor to Orchestrator: Companies are no longer just buying transportation; they are outsourcing the management of their entire supply ecosystem. 3PLs integrate all parts of the supply chain, managing a complex web of partners, including small regional carriers and customs brokers. **Shift from Physical Assets to Predictive Algorithms:** While the traditional advantage in logistics is thought to be owning a fleet of trucks, ships and warehouses, there is now a more powerful differentiator - data and predictive intelligence. This means a 3PL's biggest asset is now its technology stack, helping it to optimise inventory placement, and forecast disruptions before they occur.

VALUE CHAIN

Inbound Logistics →	Value-Addition →	Order Management →	Outbound Transport →	Reverse Logistics
Shipments from the client's suppliers are stored as inventory in warehouses.	It involves performing customization and quality checks before shipping to customers.	Retrieving items from their storage locations, packing and labelling before getting it shipped.	Includes carrier selection based on route and speed, dispatch, tracking and last-mile delivery.	Handling returns from customers, inspecting the returned items and processing refunds.

KEY NUMBERS





The Global 3PL market is expected to **1.5× in 5 years**

BUSINESS MODEL

3PL is built on economies of scale, by each 3PL investing in heavily capital-intensive assets like warehouses, advanced sorting machinery and fleets of trucks; too expensive for most individual businesses to own and operate. This becomes an advantage and enables flexibility by helping them instantly scale up without finding and leasing a new space during a peak season.

Transactional pricing (pay-per-use) is the most common pricing model, in which the client is billed for specific quantifiable actions. Cost-plus pricing is a model where the client is charged for the actual costs incurred and a percentage markup as management fees. Gainsharing is a model in which a percentage of the cost-savings of the client are received by the 3PL.

CHALLENGES

Margin Erosion: Services getting commoditised to fulfill client needs causing price competition.

Complex Cost Structures: Lack of transparency in the billing process leading to damaged client trust.

Rising Operating Costs: Fuel price volatility, scarcity of warehousing space and rising real estate prices.

Last-mile Delivery Complexity: Demand for same-day delivery and route inefficiency in dense areas.

Systemic Bottlenecks: Infrastructure limitations such as port congestion and carrier constraints.

MARKET DYNAMICS

Companies mainly compete on price, service quality, geographic coverage and technological capabilities.

It is easy for players to enter the industry due to the moderate barriers to entry, but it is quite challenging to scale operations.

Companies are also increasing their technological capabilities such as tracking, automation, warehouse management, etc.

Competition has gradually been intensifying among the top companies due to increasing technological offerings.

Example: C.H. Robinson's Navisphere Technology Platform, which provides the most data on shipments and carriers.

3PL

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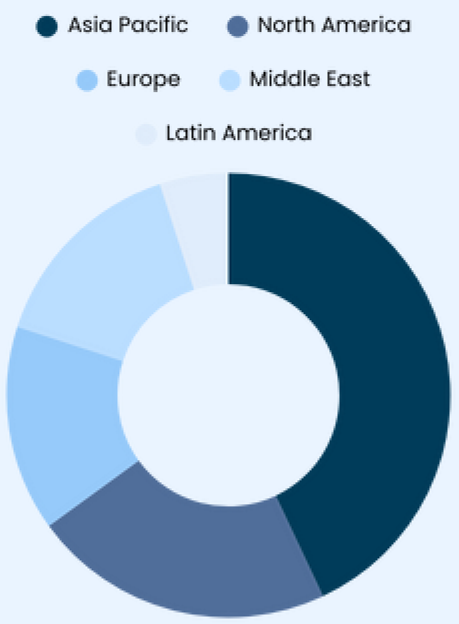
COMPETITIVE LANDSCAPE



Amazon

- *Transitioned from an e-commerce retailer to become a logistics powerhouse.*
- *Fulfillment by Amazon (FBA) enables merchants to outsource the order fulfillment process to Amazon.*
- *Amazon's retail platform serves as an unparalleled demand-generation engine.*

DHL Supply Chain and Global Forwarding

- *DHL integrates its world leading logistics provider, DHL Supply Chain, and its global freight management arm, DHL Global Forwarding.*
- *The global network of DHL along with these two divisions, provides services that can handle client demands end-to-end.*

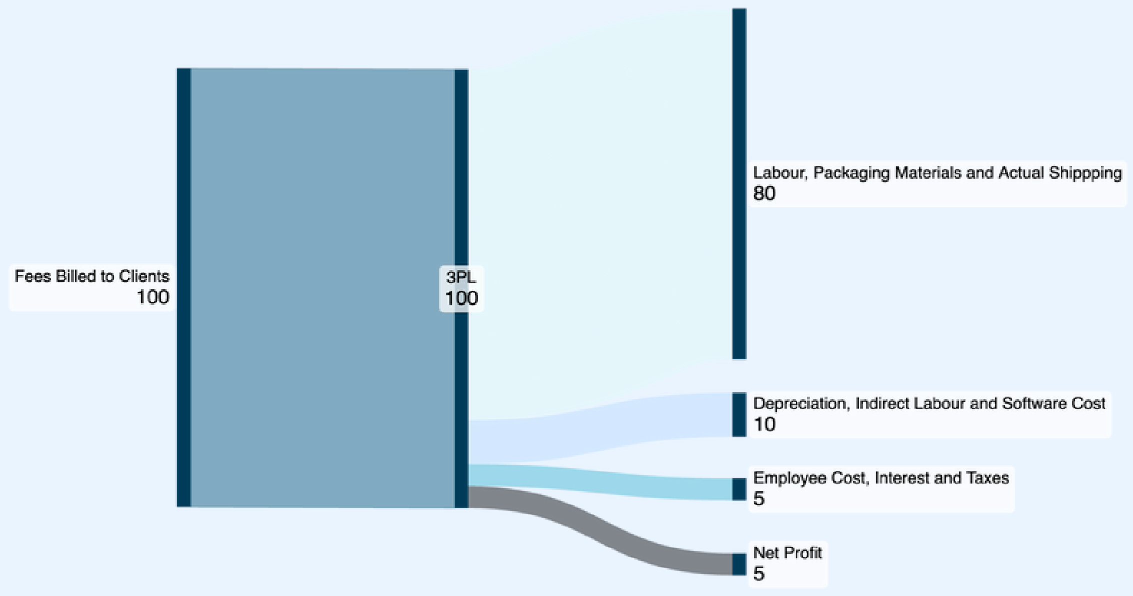
Kuehne+Nagel

- *A legacy global leader in sea and air forwarding, has positioned its contract logistics offerings to be the core of its 3PL offerings.*
- *To counteract the network advantages of its competitors, Kuehne+Nagel has made its differentiator its Digital Ecosystem, improving the technological services it provides.*

DSV

- *DSV's logistics model is asset-light and the company relies on a network of over 200,000 sub-contractors, allowing greater flexibility.*
- *Its primary growth strategy is aggressive inorganic growth through acquisitions.*
- *DSV recently acquired DB Schenker for EUR 14.3bn, making it the largest freight forwarder.*

PER-SHIPMENT ECONOMICS



ECONOMIC ENABLERS

Network & Asset Optimisation: High utilisation of warehouse space and optimised labour management are crucial. 3PLs boost profitability by aggregating freight shipments from many clients.

Value-Added Services: Services such as kitting, final assembly and reverse logistics, that go beyond traditional warehousing and transport generate higher margins.

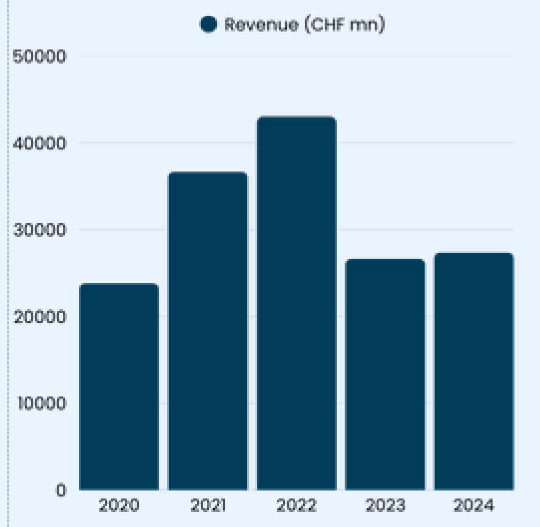
Technology and E-commerce: AI driven analytics and automation have been proven to improve service reliability and cut operating costs. The boom in e-commerce has greatly increase the demand for 3PL.

CASE STUDY: KUEHNE+NAGEL

KUEHNE+NAGEL

"Kuehne+Nagel is the third-largest 3PL provider in the world. The company has seen earnings normalise post-pandemic but is improving its future prospects via Roadmap 2026."

REVENUE TREND



- **1890 - Founding:** Founded in Germany and focused on logistics and forwarding.
- **WW2 – Scale Up:** Played a major role in logistics for Nazis and scaled operations.
- **Post WW2– Global Expansion:** Shifted to Switzerland and established a global network of distribution centers.
- **Acquisitions:** North American market position improved by acquiring ReTrans.
- **Digital Pivot:** Facing disruptions from the massive scale of competitors, the focus is on digital transformation.

EXECUTION LEVERS

Technological Innovation
Max Visibility pilot with Chorus for barcode-free tracking of shipments.

Freight Dominance
The company is a global leader in sea, road, air and contract logistics.

Cusomer Facing Platforms
myKN portal offers instant quotes, shipment booking and cargo tracking.

SIGNIFICANCE

- Kuehne+Nagel has served as a model for legacy transformation.
- Its volumes are seen as a metric of global trade health due to its market position.

FINANCIAL HEALTH

- Revenue CAGR of 3.5% from 2020-2024, while net profit margins improved 1.3%.
- Operating margins stable in 9-10% range.

FUTURE OUTLOOK

- Build on Roadmap 2026 objectives towards Roadmap 2030.
- Evolving from only logistics to a data partner.
- Navigating consolidating competition will be key.

WAREHOUSING & FULFILLMENT CENTRES

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OVERVIEW

"Warehousing is the engine of speed and efficiency in supply chains"

Redefining fulfillment : Modern warehouses and fulfilment centres handle storage, picking, packing, and dispatch. Urban micro-hubs and automation (WMS, robotics, IoT) bring inventory closer to customers, cutting delivery times and boosting accuracy.

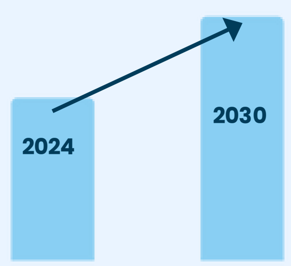
Sustainability test : The global warehousing market is ~USD 1T (2023), growing at 7–9% CAGR. India's market is ₹1.5T (2024), projected to double by 2033. Growth is led by e-commerce, pharma cold-chain, and 3PLs, while green warehouses and EV fleets tackle rising energy and real estate costs.

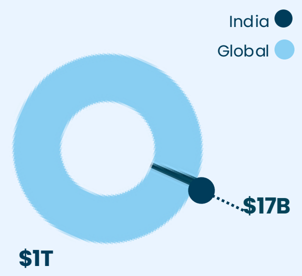
VALUE CHAIN

Inbound Logistics →	Order Processing →	Packing →	Outbound Logistics →	Reverse Logistics
Goods are received from suppliers, checked for quality, and stowed into inventory.	Customer orders are fulfilled by generating pick lists and retrieving products from storage locations.	Picked items are assembled, securely packaged, and labeled for final shipment.	Processed orders are shipped to customers from strategically located centers for faster delivery.	Management of returns involves product inspection, restocking, and handling refunds/ exchanges.

KEY NUMBERS

CAGR ~8.1%





Global vs India

The Global Warehousing market expected to reach **$1.75T** by 2030

$17B warehousing market is fueled by the 3PL and e-commerce sectors, with strong demand also driven by the automotive industry.

MODERN TECHNOLOGY

Warehouse Management System (WMS)
WMS is the core software hub providing end-to-end visibility and control over inventory, order processing, and shipping.

Automation & Robotics
Technologies like Automated guided vehicles (AGVs), robotic arms, and conveyor systems handle repetitive tasks like picking, packing, and sorting, significantly boosting speed and accuracy

IoT & Data Analytics
A network of sensors (IoT) tracks assets, monitors environmental conditions, and collects operational data. Analytics tools then use this data to optimize workflows and improve efficiency

GROWTH DRIVERS

E-commerce & 3PL Surge: E-commerce and 3PL growth fuels demand for modern fulfillment centers to manage high order volumes.

Government Policies: Government policies like the NLP are streamlining logistics, reducing costs, and attracting investment.

Automation & Tech Adoption: Adoption of automation, robotics, and WMS enhancing operational efficiency, accuracy, and speed

Manufacturing Sector Demand: Strong demand from the automotive and manufacturing sectors for storing parts and finished products.

Urban & Last-Mile Logistics: Strategic shift to urban warehouses and fulfillment hubs to enable rapid last-mile delivery.

CHALLENGES

Urban Real Estate: High costs and limited availability of real estate in urban areas challenge the setup of hubs for rapid last-mile delivery.

High Capex: Significant upfront investment is needed for land, modern construction, and advanced technology like automation and robotics.

Labor Management: Difficulty in attracting, training, and retaining a skilled workforce, compounded by rising labor costs and technological complexity.

Inventory & Order Accuracy Constant pressure to maintain precise inventory control and high order accuracy while managing fluctuating consumer demand.

Managing Reverse Logistics: Efficiently handling customer returns is a complex and costly operational challenge that can significantly impact overall profitability.

COLD CHAIN

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OVERVIEW

"From farm to pharmacy, cold chains decide what reaches us fresh, safe, and on time."

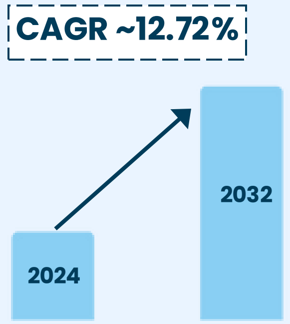
Preserves quality of Temperature-Sensitive goods: Cold Chain Logistics is about maintaining precise, low-temperature conditions throughout the supply chain—from production, storage, and transportation to distribution of temperature-sensitive goods.

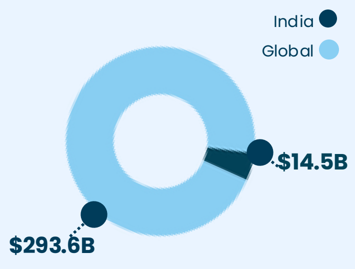
Sustainability Test: Energy-intensive refrigeration, high emissions from transport, and reliance on costly refrigerants strain margins and the environment. Long-term viability depends on green refrigerants, renewable-powered cooling, and new infrastructure, particularly in rural and semi-urban areas, to balance growth with sustainability.

VALUE CHAIN

Sourcing →	Pre-Cooling →	Cold Storage →	Reefer Transport →	Last-mile delivery
Procurement from farms, fisheries, food processors, and pharma manufacturers under strict temperature-controlled conditions.	Immediate pre-cooling, grading, packing, and treatment at source to stabilize product temperature before transit.	Refrigerated warehouses (chilled/frozen) maintain product quality, buffer inventory, and reduce spoilage risk.	Movement via reefer trucks, rail reefers, air freight, or reefer containers, maintaining continuous temperature integrity .	Urban cold delivery fleets supply quick-commerce, pharma, hospitals, and retail, ensuring temperature compliance till the final handoff.

KEY NUMBERS





Global vs India

The cold chain market is expected to **3x in the next decade.**

90% of India's **$14.5B** cold chain industry is fragmented, yet the industry is expected to **3x by 2032.**

ECONOMIC ENABLERS

Network Utilization Efficiency:	Temperature Integrity:	Integrated Storage-Transport Planning:	Digitised Control Systems:
Higher fill rates in reefer trucks and optimal warehouse throughput reduce unit costs and improve asset productivity - crucial in a high-capex segment.	Better pre-cooling, insulated packaging, and IoT monitoring reduce spoilage (currently ~30-35% loss for perishables in India) and unlock margin recovery.	Aligning harvest cycles, factory output, and distribution slots helps optimize fleet routing, cut dwell times, and stabilize yields across seasons.	WMS, IoT sensors, SCADA, and remote temperature telemetry lower manual errors, reduce compliance breaches, and increase throughput.

GROWTH DRIVERS

Rising Pharma & Vaccine Demand: Global healthcare systems rely on temperature-controlled supply chains for safe and secure distribution.

Processed Food and Beverages: F&B contributes more than 75% of the revenue, and growth of packaged food and dairy further fuels the demand.

E-commerce & Retail Expansion: Online grocery and fresh produce delivery requires robust cold logistics to meet consumer expectations.

Governmental Push: Governemts are accelerating subsidies, and infrastructure investment. In India, new projects worth ₹743 Cr are already in the pipeline.

CHALLENGES

Energy Intensity: Refrigeration consumes 40% of total operating costs, exposing operators to power shortages and rising energy prices.

High Capex: Temperature-controlled warehouses, and last-mile cold distribution requires heavy upfront investment and long payback cycles.

Loss of Produce: Weak cold storage and transport infrastructure waste one-third of total perishable output, translating into massive margin leakage

Infrastructure Imbalance: Over 70% of cold storage capacity is concentrated in a few states and cities.

Regulatory Hurdles: Strict food safety, pharma, and export standards require end-to-end temperature traceability, which is both costly and complex.

COLD CHAIN

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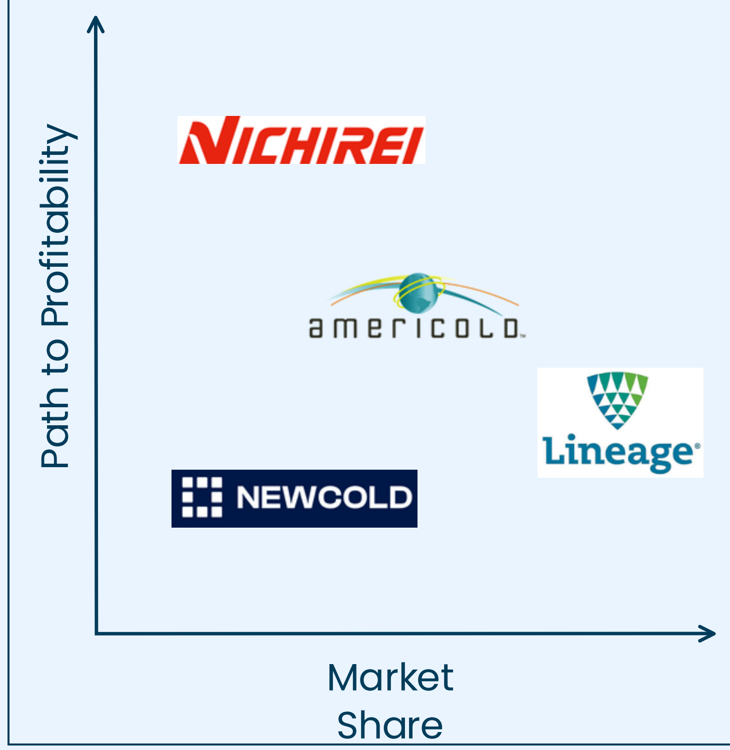
COMPETITIVE & FINANCIAL SNAPSHOT



Americold Logistics

- **Revenue:** $2.62 Bn
- **EBITDA:** $550 mN
- **Market Cap:** $3.88 Bn

"Stable revenues with slim OPM, reflecting steady but low-margin operations."

Lineage Logistics

- **Revenue:** $5.32 Bn
- **EBITDA:** $994 Mn
- **Market Cap:** $10.85 Bn

"Thin OPM (~2%) and ~25% EBITDA margin, valued ~$10.7B on growth and acquisitions."

Nichirei Corporation

- **Revenue:** $4.6 Bn
- **OPM Margin:** 5%
- **Market Cap:** $2.9 Bn

"Strong Japanese leader but market cap only ~$2.9B, signaling market discount for low-growth logistics."

NewCold

- **Revenue:** $ 564.9 Mn
- **Funding:** $100 Mn
- **Revenue driver**: long-term service contracts

"Tech-driven automated warehouses, growing rapidly in EU & US."

FAILURE POINTS IN THE COLD CHAIN

First-Mile	Cross-Docking	Extended Dwell	Last-Mile Delivery
Lack of pre-cooling at the source causes immediate spoilage risk and weakens shelf-life for the entire chain.	Transfers during loading, unloading expose products to ambient spikes, leading to irreversible quality degradation.	Long waits at ports, ICDs, airports and customs increase temperature drift and raise spoilage and energy-holding costs.	Urban heat, delivery delays and improper handling during final handover often break cold integrity and cause quality failures.

ECONOMIC ENABLERS

Network Utilization: Higher fill rates in reefer trucks and optimal warehouse throughput lower unit costs and improve asset productivity.

Tech enabled efficiency in operations: Faster inventory turnover and tech-enabled units reduce spoilage, free up capital, and support sustainable growth.

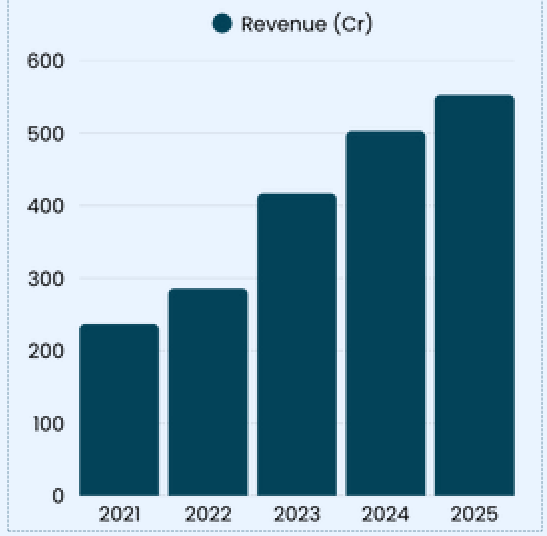
Capital Discipline: Adequate funding and prudent cost control ensure runway for scaling and progress toward breakeven.

CASE STUDY: SNOWMAN LOGISTICS

SNOWMAN

"Snowman has emerged as a key cold-chain enabler in India, delivering ₹553 Cr revenue in FY25 with improving profitability; capacity scale-up to 154k+ pallets and a 600+ vehicle fleet provides strong leverage for sustained growth."

GROWTH OUTLOOK



- **1993 - Launch:** Incorporated to provide cold-chain solutions.
- **2014 – IPO:** Listed on Indian stock exchanges.
- **2021–24 – Expansion:** Capacity scaled to ~150k pallets, 43 warehouses.
- **2025 – Scale:** 44 warehouses in 21 cities, 154k+ pallets;
- **2026E – Growth Focus:** Further Tier-2/3 city expansion & pharma-grade infra.

EXECUTION LEVERS

Fleet Strength: 296 owned and 325+ leased vehicles ensuring last-mile reach.

Pan-India Footprint: 44 warehouses across 21 cities built over 3 decades.

Multi-Sector Play Serving FMCG, QSR, pharma, dairy, poultry, seafood, and retail.

SIGNIFICANCE

- India's **largest** listed cold-chain logistics player.
- Reduces spoilage & ensures pharma compliance.

FINANCIAL HEALTH

- Positive EBITDA, and good cash-flow.
- Operating costs rising faster than revenues.
- Declining margins.

FUTURE OUTLOOK

- **Scaling** pharmaceutical storage and automation.
- **AI/IoT adoption** for efficiency.
- **Steady margin** recovery expected.

DIGITAL FREIGHT PLATFORMS

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OVERVIEW

"Freight is no longer moved just by roads, but by algorithms optimizing every mile."

A digital freight forwarder: comprehensive logistics ecosystem; connects carriers & shippers with AI, blockchain, data analytics; ensures smooth, transparent, eco-friendly transport; streamlines operations, enhances efficiency, reduces idle mileage, lowers CO_2

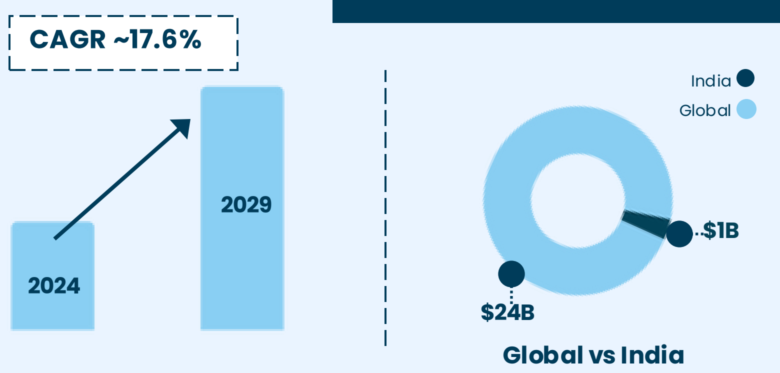
Sustainability test : Digital freight platforms face high tech costs, uneven carrier adoption, and thin margins. Long-term sustainability depends on proving CO_2 savings, cutting idle mileage, and achieving scale through dense shipper–carrier networks to unlock profitability.

VALUE CHAIN

Onboarding → **Matching** → **Tracking** → **Compliance** → **Optimization**

- **Onboarding:** Shippers post loads, carriers onboard via platform/mobile. verification ensure trust & reliable supply
- **Matching:** AI optimizes routes, cuts empty miles, sets dynamic pricing, and confirms bookings.
- **Tracking:** Real-time GPS/IoT visibility provides ETAs, alerts, and disruption management with continuous, shipment updates.
- **Compliance:** Digital contracts, invoicing, customs, and automated settlements streamline operations.
- **Optimization:** Data analytics drive forecasting, efficiency gains, and CO_2 reduction through smarter planning

Market Dynamaics



The Global digital freight platforms market is expected to grow to **$54B** by 2029

India's **$1B** digital freight platform market is led by e-commerce sectors.

How is Digital Logistic market segmented?

By Component: Solution, services

By System: Tracking and Monitoring Systems, Information Integrated Systems, Database Management Systems, fleet Management systems

By Deployment Mode: cloud, on-premises

By Application: Warehouse Management, Labour management, Transportation management

By End-User Industry: government, Aerospace, Defence, Automotive, Oil And Gas, Retail, Manufacturing, Healthcare, Other End-User Industries

GROWTH DRIVERS

Cross-Border E-commerce Boom: Retail to reach $8T by 2027, giving platforms a much stronger competitive edge in global markets and cross-border trade.

Cloud Adoption: 71% of freight platforms use cloud for scalability, compliance, and faster SME adoption across all regions with improved stability.

API Connectivity: APIs cut procurement from days to seconds, lowering costs and boosting carrier revenue on major lanes through better automation.

AI Pricing: AI sets optimized rates in milliseconds, lifting carrier revenue by 17% while cutting admin cycles and boosting fraud detection

Transparency Demand: Shippers want full visibility with audit trails and emissions reporting for faster shipment decisions and fewer disputes at every stage.

CHALLENGES

Thin Margins: Intense price competition limits profitability with margins hovering in the single digits (approx. 7-12%)

High Capex: AI and automation require heavy upfront investment, delaying profitability for newer platforms until scale.

Fragmented Market: Varying regulations and infrastructure across regions make scaling difficult and slow for digital players.

Tech integration Gaps: Legacy TMS and customs systems create integration challenges, slowing platform adoption.

Trust & Liability: Shippers require proven performance, making trust and clear liability essential for platform success.

DIGITAL FREIGHT PLATFORMS

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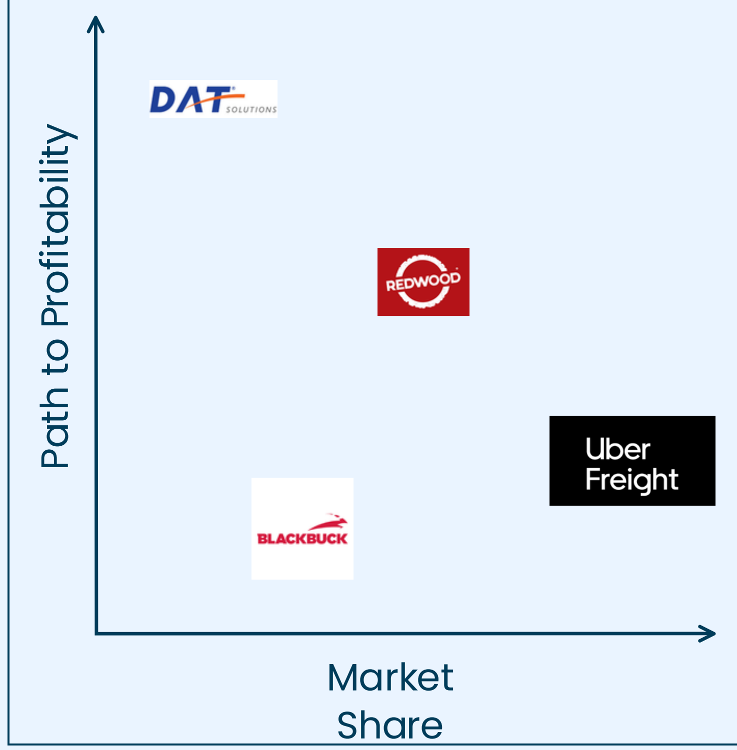
FINANCIAL



Uber Freight

- **Revenue:** $5.24B (FY 23 est.)
- **EBITDA Margin:** -5% to -10%
- **Load volume:** 3M+ | **AOV: $2000**

"Global scale, still loss-making"

DAT Solutions

- **Revenue:** $0.5B (FY 23est.)
- **EBITDA Margin:** 25-30%
- **Load posts:** 400M+ | **AOV:** subscription based

"Profitable SaaS load board, strong analytics moat"

Redwood Logistics

- **Revenue:** ~ $1.4B (FY23 est.)
- **EBITDA Margin:** 5 to 8%
- **Load Volume:** 1M+ | **AOV: ~** $1500-2000

" Hybrid 3PL + digital freight tech (RedwoodConnect)"

Blackbuck (INDIA)

- **Revenue:** $850M (FY23)
- **EBITDA Margin:** -15% to -20%
- **Loads:** 20M+ | **AOV:** ~$220-250

"India's largest digital trucking marketplace; nearing profitability in FY25"

PLATFORM VALUE CREATION

Carrier-side value: Higher truck utilization **(up 20-30%)** with fewer empty miles, faster payments improving cash flow, and steady access to loads. Platforms also reduces admin work and offer better route planning.

Shipper-side Value: Lower costs **(8-15%)** through better and reduced empty miles, along with transparent end-to -end shipment tracking, Digital platforms also ensure more reliable service quality with consistent delivery performance and faster issue resolution

Platform Value: Recurring revenue with high user-retention, supported by strong network effects. A sticky multi-service ecosystem(tracking, insurance, invoicing) Keeps shippers and carriers consistently engaged.

FREIGHT EFFICIENCY LEVERS

Dynamic Routing & load Matching: Real-time consolidation reduces empty miles and raises asset utilization

Backhaul Optimization : Platforms fill return loads, reducing cost per km by 20-30%

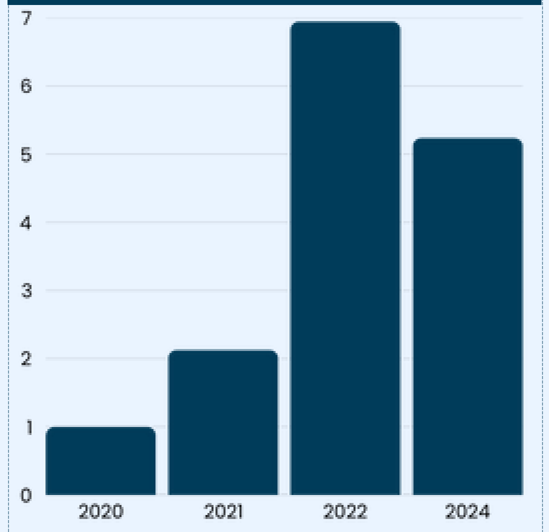
Multi-leg planning: combining smaller shipments->fewer trucks ->higher earning per km

CASE STUDY: Uber Freight

Uber Freight

"Uber Freight is one of the largest digital freight platforms globally, but despite ~$6.5B in revenue, it remains loss-making, pressured by weak freight demand and overcapacity"

GROWTH OUTLOOK



- **2021 - Acquisition:** Acquired Transplace for $2.25B, expanding into managed transportation & 3PL
- **2022 -Scale Up:** Revenue crossed $6.95B (+42% YoY), driven by integration of Transplace
- **2023 - Market softening:** Revenue fell to $6.45B (-7% YoY) as freight demand slowed
- **2024 - Efficiency Focus:** Cost cuts, automation, and AI-driven procurement; path to breakeven pushed forward
- **2025E - Stabilization:** Expected recovery as freight rates normalize; breakeven targeted by FY26

EXECUTION LEVERS

Platform+ Network
Combines digital brokerage with managed transportation (via Transplace)

Tech & AI
Predictive pricing, automated load-matching, real-time tracking for shippers

Enterprise Focus
Serves 1000+ large enterprise shippers, including Nestlé, AB InBev, and Mondelez.

SIGNIFICANCE

- Among the largest digital freight brokers globally
- Pioneer in tech driven load matching
- **Bridges Gap** between digital brokerage and traditional 3PLs

FINANCIAL HEALTH

- Revenue(2023): $5.24B(**-7% YoY**)
- EBITDA Margin: -5% to -10%(**loss making**)
- Gross Bookings:~ $18B
- Scale: 3M+ annual loads handled

FUTURE OUTLOOK

- **Expanding** AI-driven automation in procurement& routing
- growth in contract freight+ managed services
- Expected breakeven in 2-3 years as freight market stabilizes

LAST MILE DELIVERY

OVERVIEW

"The last mile is where the future of e-commerce is delivered."

Definition & Scope : Last-mile delivery refers to the final stage of the logistics chain where goods are transported from the distribution center, hub, or local warehouse to the customer's doorstep. it is the most complex and cost-intensive segment of logistics, requiring speed, flexibility, and high service reliability.

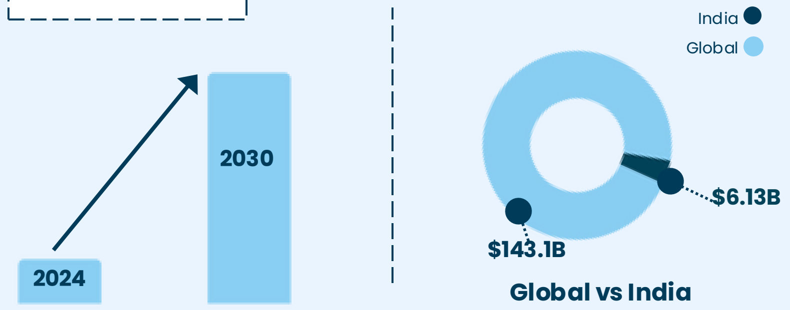
The Industry Impact: Last-mile delivery accounts for nearly 40-50% of total logistics costs, driven by e-commerce expansion, rising customer expectations, urban congestion, and the need for rapid fulfillment. innovations such as micro-fulfillment centers, gig-workforce, autonomous delivery, and hyperlocal logistics are reshaping the industry.

VALUE CHAIN

Order Processing & Entry	Hub Inbound & staging	Route Assignment	Scanning & Loading	Final Delivery and Proof
Orders enter a centralized system, giving both sender and recipient real-time tracking.	Orders reach the hub, are unloaded and prepared for the last mile delivery	Packages are sorted and assigned to delivery agents using optimized routes.	Orders are scanned and loaded onto vehicles to ensure accurate tracking	Packages reach the customer, and proof of delivery is recorded(via signature, photo etc.)

KEY NUMBERS

CAGR ~8.8%



The Global last mile-delivery is expected to grow to **USD 258.68 B**

North America Market Revenue share, 2022 is **31%**
Asia pacific is the Fastest growing market

COMPETITIVE LANDSCAPE

AMERICAS
amazon Global Logistics, ups, FedEx Express, United States Postal Service

EUROPE
dpd, EVRi, Royal Mail, GLS. Parcels to People

ASIA
DELHIVERY, Ecom Express, SF Express, YAMATO

MIDDLE EAST & AFRICA
aramex, emirates post, fetchr

GROWTH DRIVERS

E-commerce Boom: Explosive online shopping growth, especially in Tier2–3 cities, is driving higher parcel volumes and denser last-mile networks.

Quick Commerce Expansion: Demand for 10-30 minute deliveries is pushing hyperlocal warehousing, rider fleets, and high-frequency delivery models.

Urban Micro-Fulfillment: Dark stores, cloud warehouse, and neighborhood hubs reduce delivery distances and enable faster same-day/next-day fulfillment.

Fleet Diversification: Growth of EVs, e-bikes, and cargo scooters lowers operating costs, reduce emissions, and supports sustainable last-mile operations.

Payment & COD Digitization: increased digital payments and reduced COD friction enhance delivery success rates and improve cash cycles.

CHALLENGES

High Delivery Costs: Labor cost, fuel, failed deliveries, returns, and urban congestion increase per-order cost

Traffic & congestion: Urban areas face bottlenecks, slowing down delivery times. Dense traffic often disrupts planned routes

Labor Shortages: High churn among gig workers and rising demand for trained drivers. this creates persistent workforce shortages

Failed Deliveries & Return: failed deliveries increases operational costs due to re-attempts and complicates the management.

Service & Visibility Issues: Inconsistent tracking, delays, and limited visibility reduce service quality and customer confidence.

LAST MILE DELIVERY

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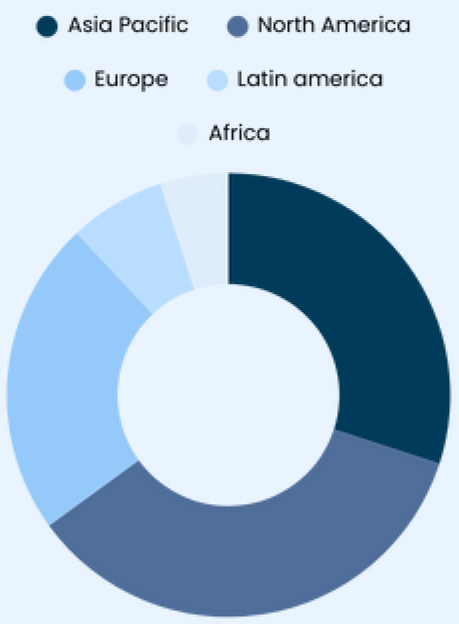
COMPETITIVE LANDSCAPE



Integrators/3PLs

Ex: DHL, EcomExpress, Shadowfax, XpressBees

- *Handle B2C e-commerce, hyperlocal, and same-day delivery.*
- *Strengths in route optimization, larger rider networks, and COD management.*
- *High serviceability across Tier- ⅔ Indian citites.*

Hyperlocal/ Quick Commerce

Ex: Zepto, Blinkit, SwiggyInstamart, Dunzo

- *operate dark stores & 10-30 min delivery windows.*
- *optimize micro-fulfillment, batching, and dense delivery networks.*
- *Focus on perishables, grocery, household essentials.*

Courier/Parcel Networks

Ex: DTDC, Blue Dart, India Post

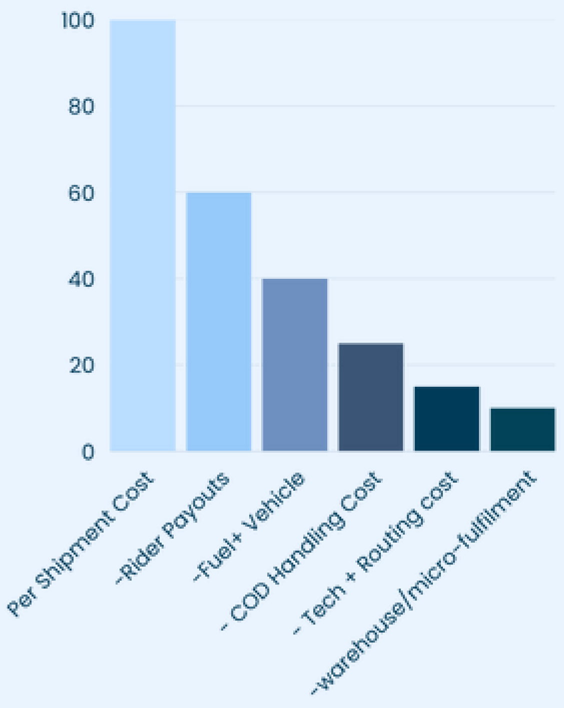
- *Strong for documents, parcels, and SME shipments.*
- *High reliability; wide pin-code coverage*
- *Compete on speed, brand trust, and enterprise contracts.*

Retailer-Owned Delivery

Ex: Amazon Logistics, Flipkart Last mile, jiomart delivery

- *Fully integrated from marketplace to doorstep.*
- *Enable Prime, Next-Day, and Same-Day delivery.*
- *Focus on cost-efficiency via automation and routing algorithms.*

PER-SHIPMENT ECONOMICS



Cost Pressures :
Last-mile operations make up 50-55% of total logistics cost, driven by high rider payouts, fuel, vehicle maintenance, and urban congestion. Failed deliveries, reattempts, RTO add another 10-15% to the cost per shipment.

Path to Breakeven:
Profitability relies on maximizing drops per rider per hour, reducing empty miles, and ensuring high-density delivery clusters .
Higher density=lower cost drop(grocery, food, quick-commerce
lower density= higher cost per drop(B2C long-distance parcels)

ECONOMIC ENABLERS

Load Factor Efficiency: Higher payload utilization across freighters and passenger belly space reduces unit costs and improves profitability. Airlines target 75–85% load factors to keep operations viable.

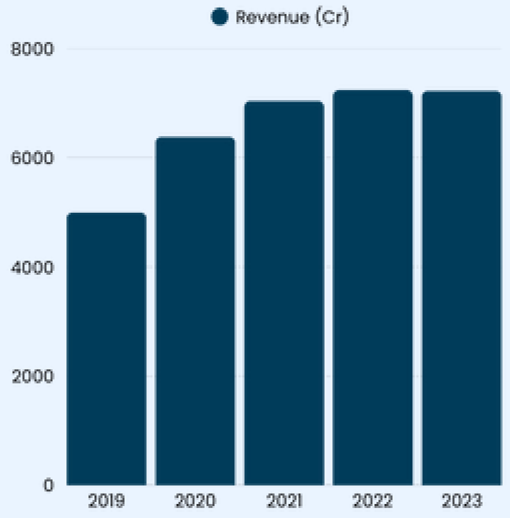
Cargo Mix Value: Premium cargo (pharma, electronics) yields 20–30% higher revenues than general cargo, strengthening margins and reducing exposure to price swings.

Policy & Infrastructure Support: Schemes like Krishi UDAN 2.0 and airport cargo hubs cut operating charges and boost regional connectivity, supporting volume growth in India's market

CASE STUDY: DELHIVERY

"Delhivery is India's largest independent last-mile logistics provider, enabling e-commerce, D2C brands, SMEs, and enterprises with nationwide coverage and strong tech, infrastructure, and automation"

REVENUE TREND



- **2011 - Founding:** Launches as a hyperlocal delivery startup in Delhi NCR; focuses on last-mile B2C shipments
- **2013-E-commerce Expansion:** Partners with major marketplaces; scales parcel delivery operations across India.
- **2021-Automation Drive:** Deploys automated sorters, routing engines, and real-time visibility platforms.
- **2021- Network Scale-up:** Crosses 18000+ pin codes, integrates trucking terminals.
- **2023-Infrastructure Growth:** Expands mega-gateway capacity; enhances Part-truckload(PTL) and end-to-end supply chain services.

EXECUTION LEVERS

Network Density & Coverage
18000+ pin codes served
Automated sortation centers, strong delivery partner ecosystem

Express Parcel Expertise
core revenue driver, Robust RTO management, strong for e-commerce, D2C, and SME segments

Tech & Ops
ML-based route optimization, Real-time tracking & SLA prediction, Automated line-haul planning.

SIGNIFICANCE

- India's largest independent last-mile logistics
- Backbone for e-commerce and emerging D2C brands
- scales delivery for tier-2 /3 India

FINANCIAL HEALTH

- FY23 revenue: ₹7,225 Cr
- EBIDTA improving with network optimization.
- Handles 3.5M+ shipments per day
- largest-scale automation

FUTURE OUTLOOK

- Expand EV-based last-mile fleet; deepen sustainability initiatives
- Increase automation in hubs & micro-fulfilment
- Scale PTL & integrated supply chain solutions.

QUICK COMMERCE

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OVERVIEW

"10-minute delivery is fundamentally rewriting India's retail playbook."

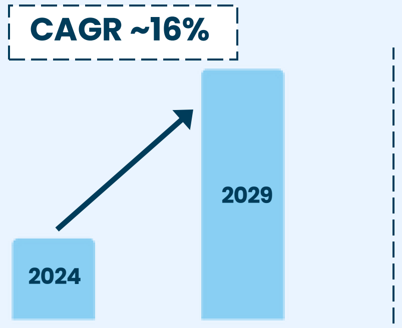
Redefining retail : Quick commerce delivers groceries, meals, and essentials in 10–30 minutes through dark-store networks. It has evolved from traditional e-commerce to meet the demands of urban, mobile-first millennials and Gen Z, who now account for over 65% of India's e-grocery orders.

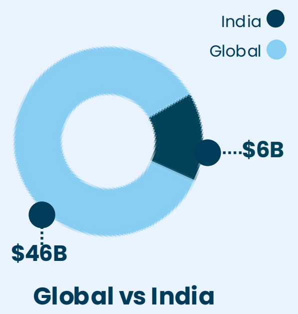
Sustainability test : The model thrives on high-frequency, low-basket transactions, but margins are pressured by fulfillment costs and weak unit economics. Long-term success will hinge on denser order clusters, higher average order values, and more efficient operations to make growth sustainable.

VALUE CHAIN

Sourcing	Fulfilment Ops	Delivery Fleet	User Interface	Reverse Logistics
Proximity-led inventory through dense dark-store networks, with 20K sites covering 2-3 km radii.	Rapid picking and dispatch cycles, averaging under five minutes across high-volume stores.	Hybrid rider model enabling sub-30-minute reach, supported by 150K+ active delivery partners.	Mobile-first platforms with real-time tracking, tapping into India's 624M+ smartphone users.	Returns and refunds managed at ~8–10% of orders, a critical challenge in thin-margin models.

KEY NUMBERS





The Global Q-commerce market expected to **4× in 5 years**

India's **$6B** Q-commerce market is grocery led, with meals and essentials emerging.

CONSUMER FUNNEL

Awareness: India's **624M** smartphone users and **65% <35 yrs** drive adoption. Urban, mobile-first lifestyles fuel demand for 10–30 min delivery.

Engagement: Active users average **6 orders/month** in 2024 (+40% YoY). Usage peaks at lunch and evening hours. High frequency drives growth but adds cost pressure.

Conversion: **₹400–600** AOV, with fresh produce at 35% of baskets. Frequent low-ticket orders strain profitability. Viability hinges on density and basket size.

GROWTH DRIVERS

Urbanization: 35%+ of Indians live in cities, driving demand for hyperlocal, convenience-led delivery.

Youth Demographic: 65% of the population is under 35, fueling adoption of app-based quick commerce.

Smartphone Penetration: 624M+ users make mobile-first platforms the natural entry point for Q-commerce.

E-grocery Surge: Is evident since over 65% of India's online grocery orders now fulfilled via Q-commerce.

Funding Momentum: Leading players raising $1.5B+ in 2024 to expand dark stores and scale operations.

CHALLENGES

Unit Economics: Thin margins persist as frequent low-ticket orders make it difficult to cover fulfillment and delivery costs.

High Capex: Scaling to 20K+ dark stores demands continuous capital for infra, inventory, and last-mile networks.

Workforce Dependence: Reliance on 150K+ gig riders raises concerns around churn, compliance, and rising labor regulation.

Returns Burden: With 8–10% of orders refunded, losses compound quickly in a model already operating under pressure.

Regulatory Hurdles: FDI restrictions, complex GST rules, and evolving gig-worker policies create ongoing uncertainty.

QUICK COMMERCE

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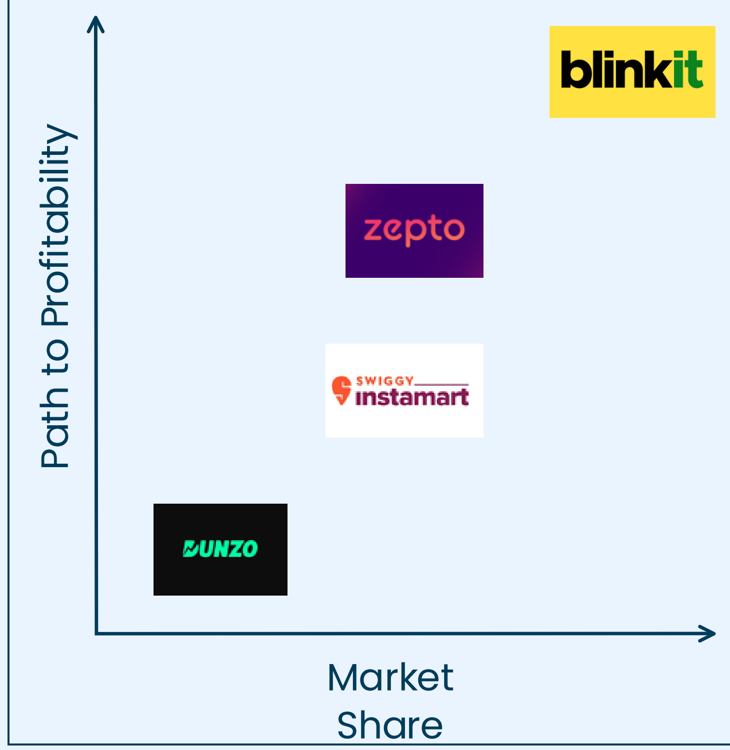
COMPETITIVE & FINANCIAL SNAPSHOT



Blinkit

- **Revenue:** ₹2,285 Cr (+119% YoY)
- **EBITDA Margin:** -43%
- **Orders:** 190M+ | **AOV:** ₹339

"Loss per order cut sharply from -₹25 to -₹3.5 , breakeven expected in 4-6Q."

Zepto

- **Revenue:** ₹2,100 Cr (FY 24 est.)
- **EBITDA Margin:** -30% to -35%
- **Orders:** 25M+ | **AOV:** ₹430

"Strong AOV cushions margins; fast growth, VC-backed runway."

Instamart

- **Revenue:** ~₹2,000 Cr (FY23 est.)
- **EBITDA Margin:** -45% to -50%
- **Orders:** 15-18M/month | **AOV:** ₹380

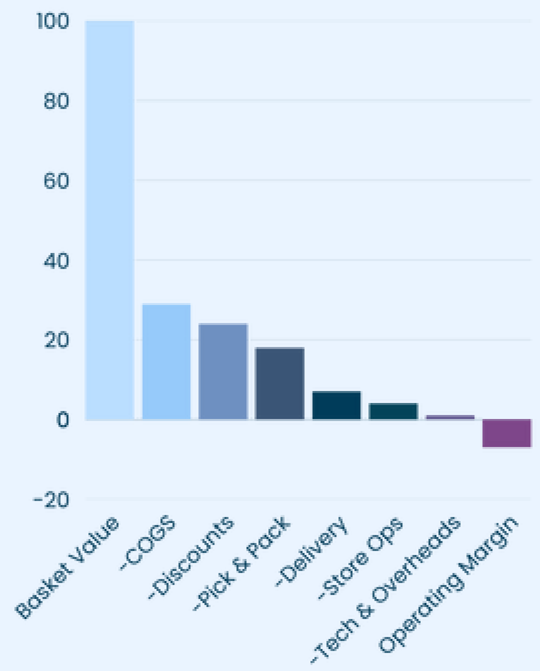
" Scale advantage via shared fleet; heavy discounting drags economics."

Dunzo

- **Revenue:** ₹225 Cr
- **EBITDA Margin:** -700%
- **Orders:** <4M/month

"Early pioneer; near-insolvent, shifting to B2B micro-logistics."

PER-ORDER ECONOMICS



Per-Order Cost Pressures : With baskets averaging ₹400–600, margins are squeezed by high fixed costs. COGS alone takes 70%+, while delivery adds 10–12%, falling only with dense volumes.

Path to Breakeven: Sustainability depends on bigger baskets, lower discounts, and higher density. Even so, margins remain negative at -4% to -10% /order.

ECONOMIC ENABLERS

Order Density: Higher orders per dark store and optimized rider routes reduce delivery costs and improve utilization.

Average Order Value : Larger baskets and repeat usage spread CAC, strengthen margins, and drive long-term stickiness.

Capital Discipline: Adequate funding and prudent cost control ensure runway for scaling and progress toward breakeven.

CASE STUDY: BLINKIT

blinkit

"Blinkit dominates India's q-commerce, but margins remain negative (-43% EBITDA). Loss per order reduced from -₹25 to -₹5.3 in FY23, signaling path to breakeven in 4-6 quarters."

GROWTH OUTLOOK

- **2021 - Acquisition:** Zomato acquires Blinkit (then Grofers) for $568M.
- **2022 - High Burn:** Loss/order at ₹25 amid rapid dark store expansion.
- **2023 - Scale Up:** Revenue ₹2,285 Cr (+119% YoY), 400+ stores live.
- **2024 - Efficiency Gains:** Loss/order narrows to –₹5.3; 190M+ orders.
- **2026E - Breakeven:** EBITDA expected to turn positive in 4-6 quarters.

EXECUTION LEVERS

Integration with Zomato
Shared fleet + customer base reduces CAC and speeds delivery.

Density Play
Urban clustering drives higher order volumes and lower per-order costs.

Tech & Ops
AI led demand forecasting and 5-min fulfillment cycles boost efficiency.

SIGNIFICANCE

- Anchors India's Quick commerce.
- Loss per order from -₹25 to -₹5.3.
- **Diversifies** Zomato's play.

FINANCIAL HEALTH

- ₹2,285 Cr Revenue (**119% increase YoY**)
- -43% EBITDA Margin
- 190M+ Orders (FY23)

FUTURE OUTLOOK

- **Expanding** into Tier-2/3 cities.
- **AI** boosts demand & supply.
- Breakeven in 4-6 quarters.

REVERSE LOGISTICS

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OVERVIEW

"Reverse logistics is redefining post-purchase in India, converting every return into a lever for retention and revenue."

Reverse logistics refers to the movement of goods from customers back to the seller or the manufacturer. Worldwide returns were about **1.8 trillion dollars** in the year 2022. This amount has more than doubled in less than ten years because of the rapid growth of online shopping.

It helps companies cut costs, recover value, reduce risks, and complete the product life cycle in a sustainable way. It also supports the circular economy by promoting reuse and responsible recycling.

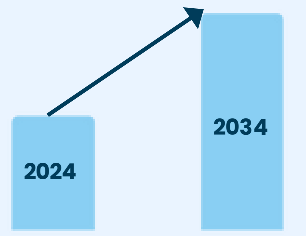
VALUE CHAIN

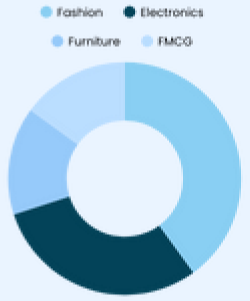
End Consumer → Return Shipping → Return Processing → Disposition → Recycle and Resale

End Consumer	Return Shipping	Return Processing	Disposition	Recycle and Resale
Customer initiates a return due to damage, dissatisfaction, expiry, or product replacement.	The product is collected or dropped off and transported back through the logistics network.	Items are inspected, verified, and sorted based on condition, warranty, and reason for return.	Returned goods are directed to the appropriate recovery path — repair, refurbish, recycle, or resell.	Products are repaired, or recycled to recover value, and are reintroduced into the market.

KEY NUMBERS

Global CAGR ~7%





Category wise Return rates

- The Indian reverse logistics market is estimated to be growing at **15-20% CAGR through 2030.**
- Processing a return costs up to **1.5x more** than the original delivery.
- Nearly **one in five** e-commerce orders is returned in India.

CLASSIFICATION

TYPES

Returns Managemnet	Handles product returns efficiently to maintain customer satisfaction.
Refurbishment	Repairs or refurbishes returned products for resale and value recovery.
Recycling (End of Life)	Recycles or safely disposes products that can't be reused.
Unsold Goods management	Manages return of unsold inventory from retailers for reuse or resale.
Packaging and Repairs	Reuses packaging and repairs defective items to reduce waste and cost.

MARKET DYNAMICS

Highly fragmented and competitive: Mix of large global 3PLs, regional/niche operators and in-house retail/e-commerce solutions.

High Barriers to entry: High setup costs and long gestation periods. Most companies operating downstream, tend to also operate upstream leaving no room for entry for new players.

High Bargaining powers of customers: Big and powerful retailers (Amazon, Flipkart, Ajio, Myntra, Pharma giants) often dictate contract terms.

High customization: Reverse logistics solutions vary greatly by industry. Electronics, retail, automotive, food or pharma all require different reverse logistic solutions.

Low Bargaining Power of Suppliers: Suppliers in reverse logistics have low bargaining power due to a fragmented market with easy substitutability.

CHALLENGES

Unpredictability: The true condition and verified availability of returned goods often remain uncertain.

COST

Cost Spiral: Handling, inspection, repackaging, and redistribution cost more than forward delivery.

Fraud & Abuse: COD refusals, fake returns, and product swaps inflate losses.

Compliance Pressure: Extended Producer Responsibility (EPR) rules on electronics, plastics, and packaging.

Customer Expectations: Refunds must be fast, or brand loyalty suffers.

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THANK YOU

For any queries, feedback or suggestions, write to us at
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